

# Presenter Guide — W3D5 · The Gammas

**What this is.** A per-slide guide so whoever presents a slide has the full context: concepts from scratch, how to read each figure, how to interpret each number, what to say (with a ready delivery script), and what questions to expect. One block per slide. **Each slide also carries a 🔍 In-depth clarifications block** — the questions that came up while reviewing the deck, resolved for the hard Q&A. The results slides (5 and 6) also carry a 🗣️ Full command block with an extended script, numbers to memorize, and trap questions with ready answers. **A note on how to use it.** I put this together from our storyboard, Andrea's 23 Jul review and our Discord discussion — it's a shared aid, not a script anyone has to follow word-for-word. Please correct anything I got wrong or remembered badly. Numbers are all checked against `pipeline/02`. **Source of truth:** `manuscript/slides/final-storyboard-5-plus-4.md` and `pipeline/02_canonical_analysis_and_slides.ipynb`. On-screen copy is English; the talk is in English.

## Coherence with the live deck — read first

This guide describes the **target** state of the storyboard (after Andrea's notes). As of 23 Jul 17:42 the live deck (19 slides) still had these edits **pending**; if you open Google Slides and see the old version, this is what's missing:

- **Slide 3:** larger question, predictions one size smaller, + the Avery 2020 anchor line.
- **Slide 4:** the 3-definition band, cut acquisition detail, replace 3 text boxes with the matrix figure.
- **Slide 6:** new title *"Two checks that narrowed what we can claim"*, 4 caveats → 2 larger ones.
- **Slide 10:** fourth test → fifth test (ST-GNN).

If we present before applying these, follow **the projected deck**, not this guide, at those points.

## Assignments (23 Jul, Zoom + Discord)

| Slide              | Presenter                            | Status                                                                                                                                  |
|--------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 3 · Intro          | Valeria                              | Confirmed (asked to go first)                                                                                                           |
| 4 · Method         | Goutham                              | Discord: "methodology is yours" — <b>conflict</b> : on Zoom it sounded like Arefeh also wanted it. Let's settle this before rehearsing. |
| 5 · Primary result | <i>open</i> (Goutham proposed Jaime) | —                                                                                                                                       |
| 6 · The turn       | <i>open</i>                          | The hinge; probably shouldn't fall to whoever's least rehearsed                                                                         |
| 7 · Conclusion     | <i>open</i>                          | Andrea approved it with no changes                                                                                                      |

Five presenters, five spoken slides (3–7), one each. Cover, team and backups take no speaking slot. With six, slide 6 splits in two (directional bridge / activation benchmark).

## Common delivery rules (apply to EVERY slide)

**From Andrea's review (she was clear about these):** 1. **Define every concept as you name it** — FC, reconfiguration, segregation. Her most repeated point; Arefeh asked for it directly. 2. **Name the axes out loud before any number** from a figure: *"always make sure people know what they're looking at."* 3. **Less text, more visual.** Trim technical detail; cut material goes to Q&A. 4. **Rehearse to the minute.** Andrea offered a rehearsal round and Friday tutorial time. Let's pre-agree who fields which questions. 5. **It's about the process, not the finding:** *"you won't get any prize if the study is better or not."* Nobody is defending a discovery; we tell what we did and what we learned.

**Language guardrails we agreed on (from Andrea + our Discord consensus) — worth avoiding on any slide:** - Not "we changed / the hypothesis evolved" → "we predicted two things; one held, one was refined". - Not "confidence interval" for the  $\pm 0.024$  → "split SD" (spread across partitions). - Not "independent external validation" → "identity-disjoint same-HCP transfer". - Not "dFC" / "dynamic FC" / "true network dynamics" → it's **static** FC aggregated per condition. - Not "model A beats model B" (scoreboard) → "specificity check". - No causality, adaptive benefit, or connectivity-specific mechanism claims.

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## Slide 1 — Cover

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**On screen:** *"Functional Connectivity Reconfiguration in N-back Working Memory"* · the five names · **The Gammas**. No figure, no speaking slot (shown while we greet).

**What to know:** The title names the **measure**, not a mechanism. "Reconfiguration" here is literally a **difference between two condition-averaged FC matrices** — no temporal dynamics implied. Andrea asked for the **team name prominent**: the deck is archived and read by next year's cohort.

**Delivery:** *"Hi, we're The Gammas. Our project is on functional connectivity reconfiguration in an N-back working-memory task."* Move to slide 2/3.

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## Slide 2 — Team

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**On screen:** *"Meet our team"* · five names · pod/TA line (Pod 884 "Ifrit Ras el Hanout" · Megapod Lotus · TA Andrea Buccellato · Project TA Azman Akhter). No science.

**Known housekeeping (not science):** - Stock photos and filler roles ("Actor/Doctor/...") — partly done; Arefeh, Valeria and Jaime have real photos, Kerem and Goutham don't yet. - **Personal email addresses** on a deck that gets archived publicly — worth finding and removing before submission.

**Delivery:** a 10-second intro, names only. Nothing more.

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## Slide 3 — Intro: the question and the two predictions · Valeria

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**On screen (verbatim):** - Title: *"Does load-related brain connectivity predict working-memory performance?"* - Anchor: *"Avery et al. 2020 predicted 2-back accuracy from task FC in the same dataset ( $r = 0.36$ ). We asked whether the change between loads carries that signal."* - *"We predicted two things"* - **Pattern** — *"A 78-feature FC fingerprint of the 2-back – 0-back change predicts performance."* - **Direction** — *"Higher load shifts networks toward integration; larger shifts accompany better performance."* - *"A distributed pattern and a one-number direction are different claims."*

**What to know (from scratch):**

*The N-back task.* Two working-memory difficulty levels: **0-back** (easy — "press when you see this specific target image"; pure attention, almost everyone is near ceiling) and **2-back** (hard — "press when the current image matches the one two positions back"; you must hold and update a buffer). 2-back is where people **differ**, so it's the condition of interest.

*The two predictions — the key point of this slide.* Not one hypothesis, two, of different natures: - **Pattern** is **multivariate and sign-free**: give the 78 numbers together to a model and see if the set predicts. It doesn't require any one of them to go up or down. Analogy: a 78-biomarker panel — you don't care about each one alone, you care whether the set is diagnostic. - **Direction** is a **single signed number**: load must **lower** segregation (more integration), and that shift alone must predict who performs better. Analogy: "fever rises, and higher fever means worse prognosis" — one variable, one direction.

Why *two and not one*. The original proposal (Arefeh, 15 Jul) was **one sentence** that already held both claims. When the data came in, one held and one didn't. Splitting them isn't rewriting history — it's making explicit what they always were: **two independent tests**. This covers us if someone asks "wasn't it a single integration hypothesis?".

Why Avery 2020. It's the calibration anchor. It predicted `acc_2bk` from task FC, **same dataset**, at  $r = 0.36$ . Cite it so that when **our 0.366** appears on slide 5, the audience judges it against that published bar, not in a vacuum. **One citation here, no more** (Andrea's instruction).

**Delivery (~1 min):**

*"Our question: does the way brain connectivity changes with memory load predict how well someone performs? A 2020 study by Avery already predicted 2-back accuracy from connectivity in this same dataset, at  $r = 0.36$  — we asked whether the change between loads carries that signal. We made two different predictions. First, a pattern: a 78-feature fingerprint of the 2-back-minus-0-back change predicts performance — this doesn't need any particular direction, just that the combined pattern carries information. Second, a direction: higher load pushes networks toward integration, and people with bigger shifts perform better — that's a single signed number. These are different kinds of claims, so we test them separately."*

**Transition:** "So how did we actually measure that? Over to the method."

**Likely questions:** - "Wasn't it one hypothesis?" → "The original proposal was one sentence containing both a pattern and a directional claim; we report them as the two separate tests they always were." - "Why 2-back, not 0-back, as the target?" → "0-back is near ceiling — everyone gets it. 2-back is where people differ, so it's where there's something to predict."

**Avoid:** saying the hypothesis changed; a second citation.

### **In-depth clarifications (presenter's questions)**

**Where do the two predictions come from? Branching or HARKing?** No branching from one "big" hypothesis, and no guessing after the fact. The original proposal (Arefeh, 15 Jul) was **one directional sentence written before the analysis** that already held two logically separable claims: that the *pattern* predicts (weak part, sign-free) and that the *direction* predicts at the individual level (strong part, signed). Splitting them is **decomposing a compound hypothesis into its two verification conditions**, not rewriting the story. What shields you from HARKing (*hypothesizing after results are known*): the uncomfortable result cuts **against** you (direction held only at group level, and activation then outpredicted the pattern). Nobody "guessing after the fact" would pick a hypothesis their own analysis deflates. Clinical anchor: a pre-specified score with two components (troponin and ECG) — reporting that one carried weight and the other didn't is following the protocol, not changing the hypothesis.

Q&A: *"The original proposal was one directional sentence written before analysis; it always contained two separable claims, and we report honestly that one held and one was refined."*

**Pattern vs. direction, objectively (not a metaphor) — the mathematical object of each:**

|                  | Pattern                                                                    | Direction                                                                    |
|------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------|
| What it is       | a vector of <b>78 numbers</b> per person, used <i>whole</i> in ridge       | <b>1 scalar</b> per person ( $\Delta$ segregation 0 – 2)                     |
| What it requires | no sign; only that the set predicts                                        | a specific sign ( $\downarrow$ segregation) <b>and</b> individual prediction |
| How it's tested  | does it predict <code>acc_2bk</code> in unseen people? $\rightarrow$ 0.366 | $\text{corr}(\Delta\text{segregation}, \text{acc\_2bk})? \rightarrow -0.105$ |
| Verdict          | <b>survives</b>                                                            | <b>refined</b> (group yes, individual no)                                    |

In one line: **multivariate-sign-free (78) vs. univariate-signed (1)**. Analogies: pattern = a 78-biomarker panel in a model; direction = "*higher fever, worse prognosis*" (one variable, one sign).

**The Avery 2020 benchmark — what it is and how you differ:** Avery et al. (2020, *J Cogn Neurosci*) predicted `acc_2bk` from task FC using **CPM (Connectome-based Predictive Modeling)**,  $r \approx 0.36$ . It's cited as a **calibration anchor**, not a rival. Two differences: 1. **Method/representation**. Avery = CPM: correlate each edge with behavior, **select** the significant ones (positive/negative), sum them, and fit a simple linear model — over a **single condition** of FC. You = **ridge over a 78-network fingerprint of a difference (2bk–0bk)** — no edge selection, and over a *contrast* between conditions, not one condition. 2. **Validation**. Avery validates in novel individuals (healthy and memory-impaired). You add the **identity-disjoint B  $\rightarrow$  A cross-cohort transfer** (0.398).

Bottom line: you match or slightly exceed Avery while predicting from the *change* between conditions, not a single condition. (The "select edges" vs. "average to networks" contrast returns on Slide 4.)

## Slide 4 — Method: from scans to a held-out prediction · Goutham

**On screen (verbatim):** - Title: "*We evaluated a 78-feature FC difference in held-out people*" - Definitions band (3 lines): - "*Functional connectivity = the Pearson correlation between the time courses of two brain regions. Each node is one of 360 Glasser ROIs.*" - "*FC reconfiguration = how much each network pair changes its coupling when memory load goes from 0-back to 2-back.*" - "*System segregation = (mean within-network FC – mean between-network FC)  $\div$  mean within-network FC, computed per participant per condition (Chan et al. 2014).*" - **1 · Cohort** — "*336 participants · HCP N-back working-memory task · 360 Glasser ROIs*" - **2 · Condition frames** — "*0-back and 2-back frames kept separate, with no temporal overlap*" - **Figure: three network matrices** (0-back FC, 2-back FC, and their difference). Caption: "*12 Cole-Anticevic networks · 12 within + 66 between = 78 values per person*" - **3 · Model** — "*StandardScaler + RidgeCV, fitted inside each training fold*" - **4 · Held-out evaluation** — "*repeated 5-fold CV · fixed holdout · 1000-permutation null · B  $\rightarrow$  A transfer (301  $\rightarrow$  100, identity-disjoint)*" - "*All splits hold out participants; '±' = split SD, not CI.*"

**What to know (from scratch) — the three definitions are yours to say out loud:**

**Functional connectivity (FC).** The **Pearson correlation** between two regions' time courses. Take how region A's signal rises and falls over time and region B's; if they move together, high positive correlation; if unrelated, near zero. Each region is a **node**; there are **360** (Glasser atlas).

**FC reconfiguration.** How much each network pair changes its coupling from 0-back to 2-back. **2-back picture minus 0-back picture.** (Analogy: a cardiac stress test — what matters is the rest  $\rightarrow$  exercise change, not either snapshot alone.)

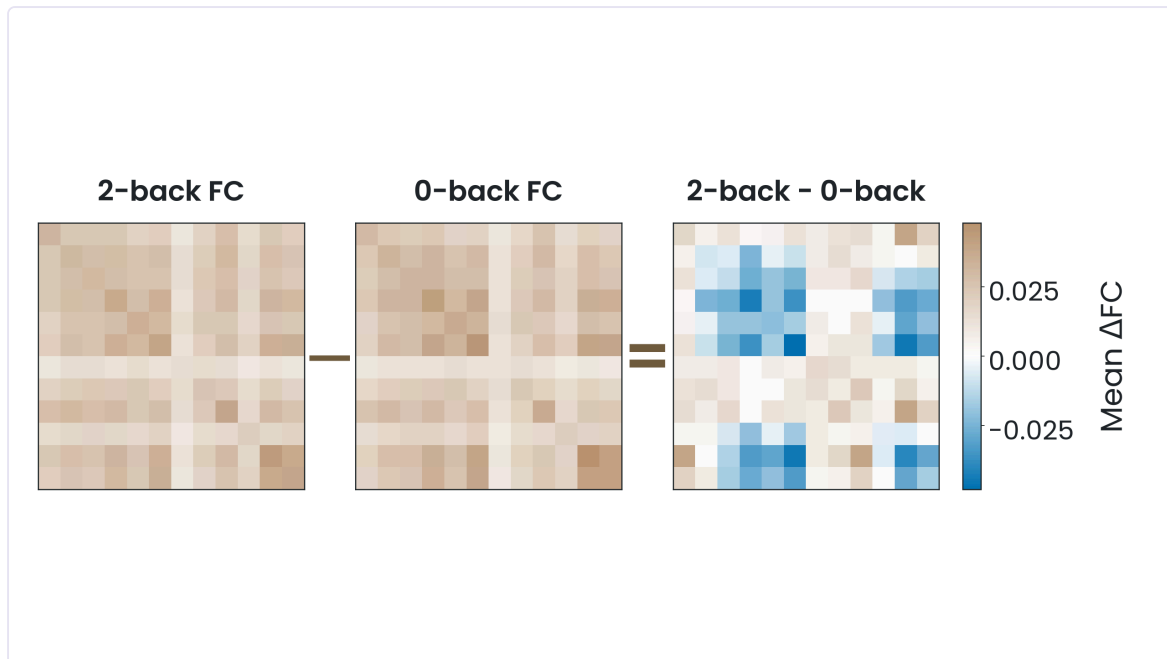
**System segregation (needs unpacking — 3 pieces):** mean **within**-network connectivity, minus mean **between**-network connectivity, divided by the within (to normalize). One number per person per condition.

**High = modular brain** (systems talking to themselves); **low = integrated** (networks mixing). You'll use this on slide 6 — remember "high modular, low integrated".

From  $360 \times 360$  to 78. The full matrices would be >64,000 pairs — impossible with 336 people without heavy overfitting. So the 360 regions are grouped into **12 networks** (Cole-Anticevic atlas: visual, motor, cognitive control, default...) and averaged: **12** within-network values + **66** between-network ( $12 \times 11/2 = 66$  unique pairs) = **78 features** per person per condition.

The model (you all know this — don't dwell): ridge with standardization, both fit **inside** each training fold — never using test info. That's the hygiene that stops CV from inflating.

The 4 evaluations (each appears later): repeated 5-fold CV  $\times 20$  · fixed holdout · 1000-permutation null · B  $\rightarrow$  A transfer with no shared identities.



**The figure (read it this way — it's not self-evident):** three  $360 \times 360$  grids. Each cell = a region pair; colour = its correlation in that condition. Scale: blue (negative)  $\rightarrow$  white ( $\approx 0$ )  $\rightarrow$  tan (positive). Panel 1 = 2-back, panel 2 = 0-back, panel 3 = the difference.

**⚠ Say this:** panels 1 and 2 share **one** scale; panel 3 has **its own**,  $\sim 10\times$  tighter. Three equally intense-looking panels do **not** mean equal magnitudes — the difference panel is stretched to make a genuinely small change visible. Only panel 3 carries a colorbar.

**Cut from the slide (available in Q&A):** TR 0.72 s, 4 s HRF shift, 312 frames per condition, 2 runs per person, the  $360 \times 360$  intermediates. Andrea: "just say it's HCP, 336 participants, this is the task. That's it."

**Exception:** the **2-runs-per-person** design does need saying if anyone asks about the open (cross-run) squares on slide 6.

**Delivery (~1 min):**

"Quick definitions first. Functional connectivity is just the Pearson correlation between two brain regions' time courses — each region is one of 360 Glasser nodes. FC reconfiguration is how much each network pair changes its coupling from 0-back to 2-back. And system segregation — we'll need it later — is within-network minus between-network connectivity, normalized: high means a modular brain, low means an integrated one. Pipeline: 336 HCP participants doing the N-back. We compute a 360-by-360 correlation matrix per condition, then summarize it into 12 within- and 66 between-network values — 78 numbers per person. Naming the axes on this figure: three network matrices — 2-back, 0-back, and their difference. The first two share one colour scale; the third has its own, about ten times tighter, so the



*difference is small even though the colours look strong. Reconfiguration is that difference. We feed it to a ridge model, fitted strictly inside each training fold, and evaluate only on held-out people."*

**Transition:** "So did this 78-feature difference actually predict anything? Yes — and it transferred."

**Likely questions:** - "Why 78 and not all pairs?" → "64,000 edges for 336 people would overfit; the network summary is the standard dimensionality reduction." - "Doesn't the task inflate correlations (coactivation)?" → "Yes, task FC can reflect coactivation, not communication — it's a stated limitation; separating them is future work (backup slide 10, test 2)." - "Acquisition parameters?" → TR 0.72 s, HRF shift 4 s, 312 frames/condition, 2 runs — ready if asked.

**Avoid:** "dynamic FC"; implying the matrices show comparable magnitudes (different scales).

### In-depth clarifications (presenter's questions)

**What is "the picture of a state"? (reconfiguration, precisely):** "The picture" = a **360×360 connectivity matrix** for one person in one condition: the Pearson correlation between the BOLD time courses of each pair of the 360 Glasser ROIs. Three precisions that kill the usual confusion: - It is **not** a raw BOLD matrix (that would be the time series, 360×~405 frames). - It is **not** a series of matrices over time (that would be *dynamic* FC — which you do **not** do). - It is a single **static** correlation matrix per condition; time is already collapsed inside the correlation.

**Reconfiguration = picture(2-back) – picture(0-back)**, cell by cell, then summarized to 78. (Cardiac stress test: the rest → exercise change matters, not either snapshot.)

**Which dataframes, train/test, and ID cleaning?** Dataset = **B ( load\_hcp )**. 339 subjects → drop **3** (81, 143, 329) with no 2-back condition → **336 analytic**. Split **by subject** (prevents leakage), seed 42 → **269 train / 67 test**; repeated 5-fold CV ×20 seeds; scaler and ridge  $\alpha$  fit **inside train only**, per fold. The external transfer (Slide 5) trains on B and tests on A **after removing the 35 overlapping identities** from B → ~301 B → 100 A, *identity- disjoint*. That's where "ID cleaning" is critical: the identity overlap between the two cohorts is removed by hand. Target is always `acc_2bk`.

**Where do the 78 features come from (out of ~64,000)?** Pure arithmetic. A 360×360 matrix → unique pairs =  $360 \times 359 / 2 = 64,620$  edges → with 336 people that's  $p \gg n$  (guaranteed overfitting). Group the 360 ROIs into **12 Cole-Anticevic networks** and average: **12** within-network values + **66** between-network ( $12 \times 11 / 2$ ) = **78**. That is, **78 = 12 + 66**.

**Cole-Anticevic: who applied it, what evidence, valid move or mistake? Does Avery do the same?** It's a **published standard atlas** (12 networks over the 360 Glasser nodes; Ji et al. 2019 partition), loaded by the official NMA/HCP framework ( `regions.npy` ) — **reused canonical material**, not an ad hoc team choice. It was a **valid move, not a mistake**: theoretically aligned with the *between-network* hypothesis (if the question is between systems, averaging to systems is *for* it), and statistically sound (78 features with  $n=336$  is  $p < n$ , guarding against overfitting). Honest nuance: averaging is **lossy** compression (it discards fine-grained edge-level signal); that's why "*matching activation/FC dimensionality*" is listed as **test 1 on Slide 10 (future work)**. **Does Avery do the same? Not exactly:** Avery controls  $p \gg n$  by **selecting edges** (CPM), you by **averaging to networks** — two standard strategies for the same problem (aggregate vs. select), with different interpretations (systems vs. specific connections).

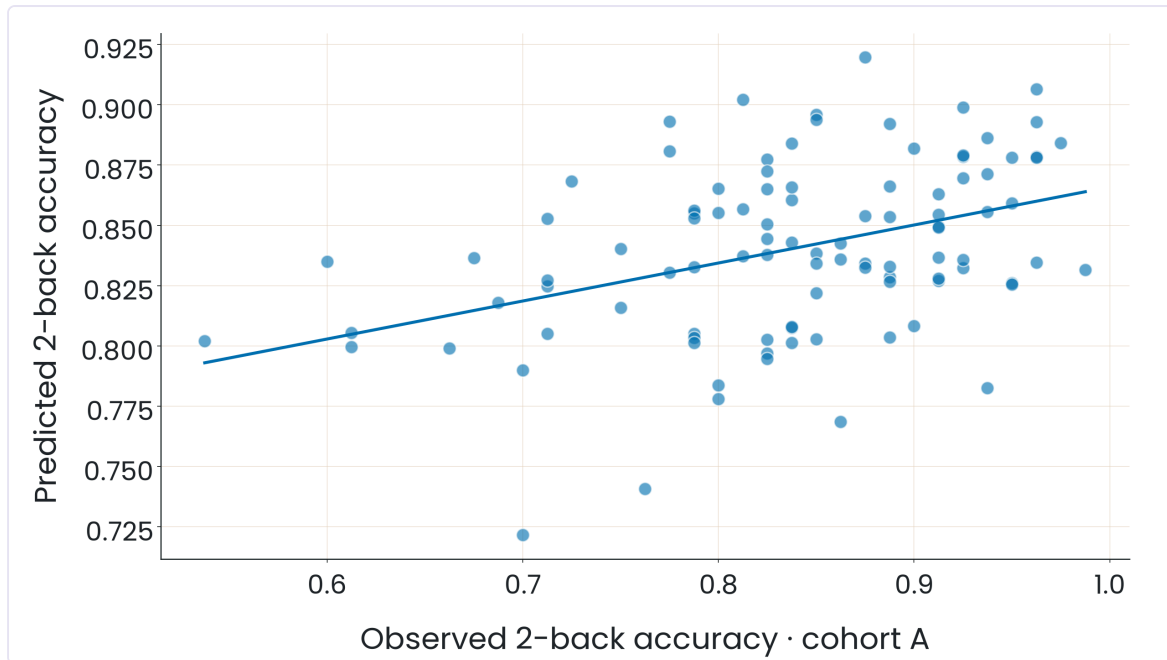
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## Slide 5 — Primary result: the pattern predicts and transfers · open

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**On screen (verbatim):** - Title: "*The FC pattern predicted performance—and transferred to a separate cohort*" - **B → A transfer** " $r = 0.398$ ; bootstrap 95% CI [0.25, 0.53]" · "301 B-only → 100 A · 35 shared identities"

removed" - **Primary repeated CV** " $r = 0.366 \pm 0.024$  · 336 participants · 78 FC features" - "Identity-disjoint same-HCP transfer—not independent-site validation." - **Figure:** scatter identity-disjoint-transfer.png .



**The figure (name the axes first):** X = each person's **actual 2-back accuracy** in **cohort A** (the test set). Y = the **predicted accuracy** from a model trained **only on cohort B** that never saw anyone in A. Each dot = one person in A. The line rises left to right: the trend. If the model were useless, a flat cloud; instead, a clear upward slope.

**What to know — the two numbers measure different things, don't conflate them:** -  $0.366 \pm 0.024$  = repeated-CV mean. The  $\pm$  is a **split SD, NOT a confidence interval**: repeat the 5-fold split 20 times, 0.366 is the mean, 0.024 is how much it moves depending on the partition. It's about **stability against partition luck**. - **0.398, 95% CI [0.25, 0.53]** = B  $\rightarrow$  A transfer. Here the model is **fixed** (trained once), so you can **bootstrap** over the 100 people in A  $\rightarrow$  this is population uncertainty in the classic sense. 35 shared identities were removed so no one is in both train and test ("identity-disjoint").

*Calibration with Avery:* 0.366 and 0.398 land right in the region of the published 0.36 — or a touch above. It's **modest in absolute terms but on par with the literature**. The Project TA's success criterion was never a high  $R^2$ , but **clearly beating chance** — which slide 9 demonstrates in detail.

⚠ It's transfer between **two cohorts of the same HCP** — stronger than reshuffling your own data, weaker than replicating at another site/scanner. Never call it independent external validation.

**Delivery (~1 min):**

"Naming the axes: horizontal is each person's actual 2-back accuracy in cohort A; vertical is the accuracy our model predicted for them — a model trained only on cohort B that never saw these people. Each dot is one person, and the trend clearly rises. Two numbers. Our primary effect is the repeated cross-validation mean,  $r = 0.366$  — and that plus-minus is a split standard deviation across partitions, not a confidence interval. Then we trained on B only and applied it to A, removing 35 shared identities:  $r = 0.398$ , with a bootstrap 95% CI of 0.25 to 0.53. This is identity-disjoint transfer within the same HCP study — not independent-site validation, but it's the same ballpark as the 0.36 Avery reported."

**Transition:** "So the prediction held. But we ran two more checks — and they narrowed what we can claim."

**Likely questions:** - "What if it's overfitting / luck?"  $\rightarrow$  backup slide 9: 1000-permutation null ( $p \approx .001$ ) + fixed holdout ( $r = 0.312$ , 67 never-seen people). - "Isn't 0.366 low?"  $\rightarrow$  "For brain-behavior prediction it's a real, publishable effect — Avery got 0.36 in the same data. The TA's bar was beating a permutation null, not a high

$R^2$ ." - "Why split SD and not a CI on 0.366?" → "The model is refit each fold, so the spread is partition sensitivity, not population uncertainty. We only claim a true CI for the fixed-model transfer."

**Avoid:** "confidence interval" for 0.366; "external validation".

### In-depth clarifications (presenter's questions)

**Do you need to say the transfer caveat, in those exact words? — Yes, always.** Yes: say it literally and don't hide it. It shows you know **how strong** the result is, and it spares you from having it raised as a gotcha. The line, projected and spoken:

*"This is an identity-disjoint transfer between two cohorts of the same HCP study. It is not a validation in a different hospital, with a different scanner, on a different population. It is stronger than just reshuffling the data you already had, but weaker than a true external replication."*

It's a **strength gradient**: reshuffle-your-own-data (weak) < transfer between cohorts of the same study (yours) < external replication at another scanner/population (strong). Saying where you land is epistemic honesty. Clinical anchor: validating a score in another subgroup of the same hospital vs. in another country — both count, but they're not the same, and saying which is which is what makes you credible.

### Full command (for the presenter — likely Jaime)

**Numbers to nail (no notes):**  $0.366 \pm 0.024$  (repeated CV,  $\pm$  = split SD) · 0.398, 95% CI [0.25, 0.53] (B → A, bootstrap) · Avery 0.36 (anchor) · backup if pushed: seed-42 null 0.405  $p \approx .001$ , fixed holdout 0.312 (67 unseen).

**The script, line by line — what each carries:** 1. *Name the axes first* (Andrea's rule): X = **actual** accuracy in A; Y = accuracy **predicted** by a model trained **only on B**. Each dot = one person in A. Upward slope = signal. 2. *The two numbers measure different things — never merge them.*  **$0.366 \pm 0.024$** : the  $\pm$  is a **split SD** (the model refits each fold → it measures partition sensitivity, not population uncertainty). **0.398 [0.25, 0.53]**: here the model is **fixed** → the bootstrap over A's 100 people is a classic CI. 3. *Close with the Avery calibration:* 0.366/0.398 land in the zone of the published 0.36 → modest in absolute terms, **on par with the literature**.

**Trap questions (with ready answers):** - "0.366 is really low, no?" → "For brain-behavior prediction it's a real, publishable effect — Avery got 0.36 in the same data. The TA's success criterion was beating a permutation null, not a high  $R^2$ ." - "Why split SD and not a CI on 0.366?" → "The model is refit every fold, so that spread is partition sensitivity. We only claim a true confidence interval for the fixed-model B → A transfer." - "Isn't predicting N-back from N-back FC circular?" → "It's a same-task association — a stated limitation. We partly control it: partialling out 0-back accuracy (general ability), the signal survives, and we anchor it in Avery's same-task design. Cleanly separating task coactivation from connectivity is future work (backup 10, test 2)." - "What if it's overfitting or luck?" → backup 9: 1000-permutation null  **$p \approx .001$**  + fixed holdout **0.312** in 67 never-seen people. - "Why train on B and test on A, not the reverse?" → "B is the larger cohort (336 vs 100), so we train on the bigger sample and test on the smaller — the stronger generalization direction."

**Delivery:** ~1 min. It's a **single-figure, two-number** slide; the risk is rushing and merging the two numbers. Breathe between "0.366 split SD" and "0.398 CI". Rehearsed transition to 6: "So the prediction held — but we ran two checks, and they narrowed what we can claim."

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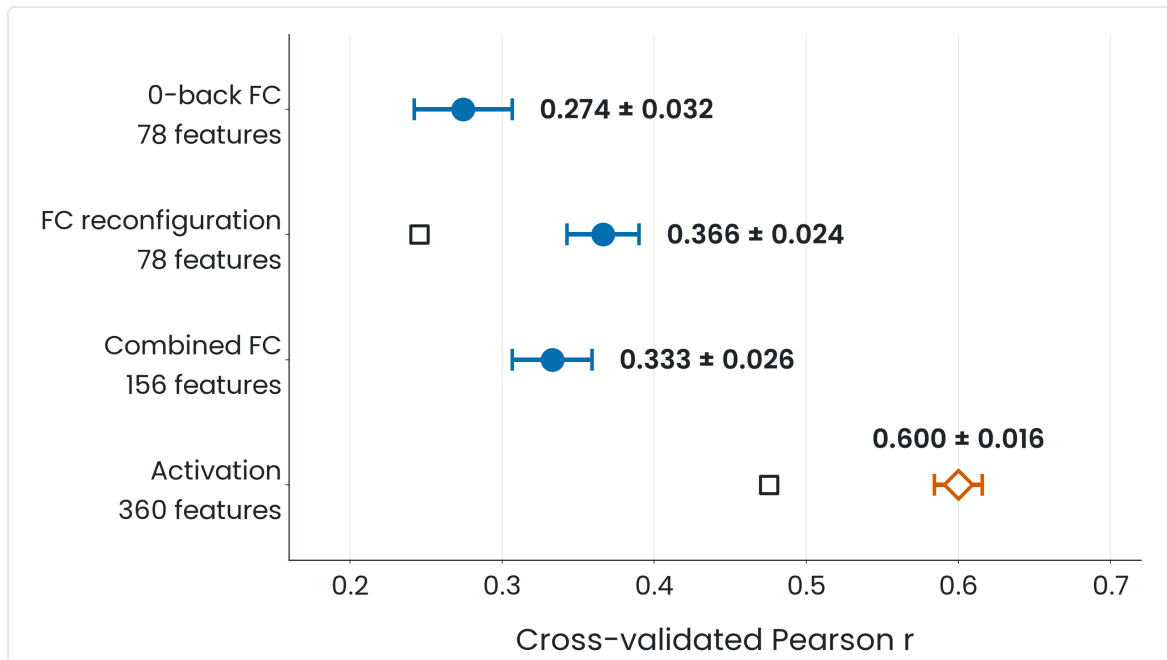
## Slide 6 — The turn: two checks that narrowed what we can claim · open (the hinge)

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**On screen (verbatim):** - Title: "Two checks that narrowed what we can claim" (retitled 23 Jul — the old one read as a scoreboard) - "Repeated-CV correlation (mean  $\pm$  split SD across 20 partitions)" - "0-back FC 0.274  $\pm$  0.032 · FC reconfiguration  $0.366 \pm 0.024$ " - "0-back + reconfiguration  $0.333 \pm 0.026$  · Activation contrast



0.600 ± 0.016" - **Direction, as predicted—but only at group level:** "Segregation fell under load (0.3271 → 0.3035;  $\Delta = -0.0236$ ;  $p = 3.45 \times 10^{-5}$ ), yet larger shifts did not predict better performance ( $r = -0.105$ ;  $p = .054$ )." - "Open squares = held-out cross-run generalization." - Two caveat lines (large): - "A specificity check, not a competition: post hoc, 360 activation features vs 78 FC features, and 0-back activation alone predicts as well (0.571), so the benchmark is not load-specific. FC reconfiguration stays the load-specific measure." - "Activation is a raw BOLD amplitude difference, not a GLM beta. Individual vascular reactivity (CVR) is uncontrolled, and this dataset carries no CVR proxy." - **Figure:** activation-robustness.png .



**Framing (say it this way):** not "we lost to activation", but "we ran **two robustness checks**, and both made us more precise". Two beats: **direction** and **specificity**.

**Check 1 — direction (answers the "Direction" prediction from slide 3):** at group level, segregation fell 0.3271 → 0.3035;  $\Delta = -0.0236$ ;  $p = 3.45 \times 10^{-5}$  (paired,  $n=336$ ) — **essentially zero by chance**. The predicted direction **exists**. But does it predict an individual? The correlation between a person's segregation change and their accuracy is  $r = -0.105$ ,  $p = 0.054$  — right at the significance border. So **no**.

**The most transferable lesson of the talk:** an overwhelming group effect does **not** guarantee individual predictive value. Just as blood pressure rises with age across a population (rock-solid) but age alone doesn't diagnose a patient. Effect size  $\neq$  individual diagnostic relevance.

**Check 2 — specificity (the figure):** dot-and-error-bar plot, horizontal, 4 rows. X = CV correlation, ~0.2 to 0.7. Each row = a different feature set, same model recipe: - **0-back FC (78)** → **0.274** - **FC reconfiguration (78)** → **0.366** (the star of slide 5) - **Combined (156)** → **0.333** — note, **lower** than reconfiguration alone: combining doesn't help, it dilutes. A hint that reconfiguration is already doing all the useful FC work. - **Activation (360)** → **0.600**, in orange/diamond — clearly above any FC.

**Open squares** over the reconfiguration and activation rows = an even stricter check (train on one run, predict on the other). By eye (**approximate, not printed numbers**): ~0.24 reconfig vs its 0.366; ~0.47 activation vs its 0.600. Honest message: cross-run replication is weaker for **both** measures.

**Why it's NOT a fair competition (say all of it):** 1. **360 vs 78 features**, unmatched → more features almost always predict more, with no biological meaning. 2. **0-back activation alone already predicts 0.571** — nearly the 0.600 of the contrast. If the person just sitting in the easy condition already gives you almost all the prediction, the signal is **not about load**; it looks like a **stable trait**. 3. **CVR (cerebral vascular reactivity):** activation is raw BOLD amplitude (not a GLM beta), influenced by vascular factors (vessel stiffness, age,

caffeine) unrelated to neural activity. This dataset can't measure or control it. FC, being a **correlation**, cancels much of that shared scale factor.

**Goutham's argument (accepted — say it if asked "why keep the weaker measure?"):** a **difference score** mathematically cancels the **stable** variance (vascular anatomy, baseline amplitude, trait-level motion) and leaves only what **changed** with the task. So it's **expected** that reconfiguration replicates worse across runs than a raw measure — not a defect that favors activation. Activation replicating better does **not** prove it's a better cognitive measure; it may be loaded with stable (vascular) noise that happens to correlate with performance. With 0-back-alone predicting 0.571, the activation advantage is driven by a **static trait**, not the task.

⚠ **Traceability gap:** the between-run reliability numbers behind this argument (reconfig  $\approx 0.024$ , activation  $\approx 0.169$ ) live in nb08, **not** in pipeline/02, so they have **no automated check**. **Don't print them on a slide** — keep them verbal, in the speaker note.

**Delivery (~1.5–2 min — the longest slide; rehearse against the clock):**

*"Two checks, both of which narrowed our claim. First, direction. At the group level, segregation fell under load — from 0.327 to 0.304,  $p$  around 3 times ten-to-the-minus-five, essentially zero by chance. So the direction we predicted is real. But does it predict individuals? The correlation between a person's segregation change and their accuracy is minus 0.105,  $p = 0.054$  — right at the border. So a rock-solid group effect does not give us individual prediction — same trap as blood pressure rising with age in a population but age alone not diagnosing a patient. Second, specificity — naming the axes: horizontal is cross-validated correlation, and each row is a different feature set. 0-back FC, 0.274. Our reconfiguration, 0.366. Both combined, 0.333 — lower, so combining doesn't help. And plain regional activation, 360 features, 0.600 — higher than any connectivity. But this is a specificity check, not a competition: it's post hoc, 360 versus 78 features unmatched, and 0-back activation alone already predicts 0.571 — so it isn't even load-specific. Activation is raw BOLD amplitude, which vascular reactivity contaminates and we can't control here. A difference score like ours mathematically removes that stable variance, so it's expected to look less reliable — that's why we keep it as our pre-specified, amplitude-independent measure."*

**Transition:** "So — what does all this add up to?" (to the conclusion).

**Likely questions:** - "So activation is better — should you use it?" → "It predicts more in this unmatched, post-hoc comparison, but it isn't load-specific and it's confounded by vascular reactivity. We report it as a specificity check, not a replacement." - "Why does the combined one drop?" → "Adding 0-back FC to reconfiguration doesn't add independent signal and slightly dilutes it — reconfiguration alone is doing the work." - "What are the squares?" → "Cross-run generalization: train on one scan run, test on the other. Both measures drop — reliability across runs is weaker than the main estimate for both." (2 runs per person — that's where this comes from.) - "Isn't this  $dFC$ ?" → "No — this is static FC aggregated per condition, then subtracted. Not dynamic connectivity."

**Avoid:** saying a model "wins"; calling reconfiguration "the most robust"; printing the nb08 reliability numbers.

### **In-depth clarifications (presenter's questions)**

**Activation vs. reconfiguration — the neuroanatomical/physiological reading (why activation predicts more).** First, what each measures, *physically*: - **Reconfiguration (FC):** a **relational** property — how the *coupling between pairs* of regions changes with load. It's a **difference** (2bk–0bk) summarized in 78. -

**Activation:** a **per-region** property — how much *mean BOLD signal* sits in each of the 360 regions, ignoring connectivity. Amplitude difference 2bk–0bk → 360 features.

Activation wins (0.600 vs 0.366) for **three stacked reasons, none of which is "connectivity is useless"**: 1. **More features (unmatched comparison):** 360 vs 78. More variables  $\approx$  more predictive capacity, no biological implication → hence *specificity check*, not competition. 2. **Reliability (psychometrics):** mean

regional activation is a **stable, reliable trait**; reconfiguration is a **subtraction of two correlations**, dominated by noise (Hedge 2018 **reliability paradox**: subtracting two reliable measures yields an unreliable difference). Less reliable → lower predictive ceiling, by measurement alone. (*The exact reliability numbers live in nb08 and are NOT printed — keep them verbal; see the traceability gap above.*) 3. **Not load-specific: 0-back activation alone** already predicts **0.571** ≈ the 0.600 of the contrast. If the person in the easy condition already gives you almost all the prediction, the signal is a **stable trait**, not working memory.

**The serious neuroanatomical caveat — CVR.** Activation here is **raw** BOLD amplitude (not a GLM beta), contaminable by **cerebral vascular reactivity**: vessel stiffness, age, caffeine — *vascular, not neural*. This dataset can't measure it. **FC, being a correlation**, cancels much of that shared scale factor. So "activation wins, connectivity is redundant" doesn't hold: activation may be winning **partly because it's contaminated with something that isn't cognition**. Clinical anchor: activation = absolute temperature of each organ (reliable, but contaminated by how warmly the patient is dressed = CVR); reconfiguration = the *change* in temperature on exercise (cancels the baseline layer, but noisier). The absolute thermometer "predicting better" doesn't mean it measures the cognitive response better.

### **Full command (for the presenter — likely Jaime)**

**It's the hinge and the longest slide (~1.5–2 min). Rehearse against the clock.** The mental frame to convey: *not "we lost to activation", but two robustness checks that forced us to sharpen.*

**Two-beat structure (say them in this order):** - **Check 1 · Direction** → answers the "Direction" prediction. Group: segregation 0.3271 → 0.3035,  $\Delta = -0.0236$ ,  $p = 3.45 \times 10^{-5}$  (real). Individual:  $\text{corr}(\Delta \text{segregation}, \text{acc}) = -0.105$ ,  $p = 0.054$  (weak). Transferable lesson: **an overwhelming group effect ≠ individual predictive value** (blood pressure rises with age across a population, but age alone doesn't diagnose). - **Check 2 · Specificity** → the 4 rows: 0-back 0.274 · reconfig 0.366 · combined 0.333 (drops!) · activation 0.600. And why it's not a fair competition (360 vs 78 features; 0-back-alone 0.571; CVR).

**Numbers to nail:** 0.274 / 0.366 / 0.333 / 0.600 · direction group  $\Delta = -0.0236$   $p = 3.45 \times 10^{-5}$  · individual  $-0.105$   $p = 0.054$  · 0-back activation alone 0.571.

**Trap questions (with ready answers):** - "So activation is better — should you use it?" → "It predicts more in this unmatched, post-hoc comparison, but it isn't load-specific and it's confounded by vascular reactivity. We report it as a specificity check, not a replacement." - "Why does the combined one (0.333) drop below reconfiguration alone (0.366)?" → "Adding 0-back FC to reconfiguration adds no independent signal and slightly dilutes it — reconfiguration alone is doing the work." - "Why keep the weaker measure in raw terms?" (Goutham's argument) → "A difference score mathematically cancels everything stable between conditions — vascular anatomy, baseline amplitude, trait motion — leaving only what changed with the task. So a subtraction is expected to replicate less across runs than a raw measure; that's not a defect. We keep reconfiguration because, by construction, it's the most amplitude-independent, load-specific measure." - "Isn't this dynamic FC?" → "No — static FC aggregated per condition, then subtracted. Not dynamic connectivity." (It's the first thing a field expert catches.) - "What are the open squares?" → cross-run (train on one run, test on the other); **both** measures drop → it comes from the **2-runs-per-person** design.

**Delivery:** the biggest risk is sounding like a scoreboard. Always anchor on "specificity check" and close with "reconfiguration stays our pre-specified, amplitude-independent measure." Transition to 7: "So — what does all this add up to?"

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## **Slide 7 — Conclusion: what we predicted and what the evidence did to it · open · keep projected in Q&A**

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**On screen (verbatim):** - Title: "Predictive signal survives; connectivity-specific mechanism remains unresolved" - **Survives — the pattern hypothesis:** "A 78-feature FC difference predicts unseen 2-back

accuracy." · "The model transfers across identity-disjoint same-HCP cohorts." - **Refined — the directional hypothesis:** "Segregation fell under load, but larger shifts did not predict better performance." · "Reconfiguration showed no clear gain beyond 0-back FC." · "FC added no clear gain over activation under the current unmatched comparison." - **Unresolved:** "Is the predictive information connectivity-specific, or shared with task activation?" · "Vascular reactivity (CVR) is not controlled in the activation benchmark." - No figure. Andrea approved it with no changes.

**What to know:** three columns that **close the loop opened on slide 3**. Survives = the Pattern prediction. Refined = the Direction prediction. Unresolved = what we honestly don't know. Naming the first two columns with slide 3's labels is what makes the story **close** rather than sound like a retraction.

**Delivery (~1 min):**

*"To close the loop we opened. Survives — the pattern hypothesis: a 78-feature FC difference predicts unseen 2-back accuracy, and it transfers across identity-disjoint cohorts. Refined — the directional hypothesis: segregation did fall under load, but bigger shifts didn't predict better performance; reconfiguration added no clear gain over single-condition FC; and FC added no clear gain over activation in this unmatched comparison. Unresolved: is the predictive information specific to connectivity, or shared with task activation? And vascular reactivity is uncontrolled. So — the predictive signal survives; the connectivity-specific mechanism remains unresolved. Thank you — happy to take questions."*

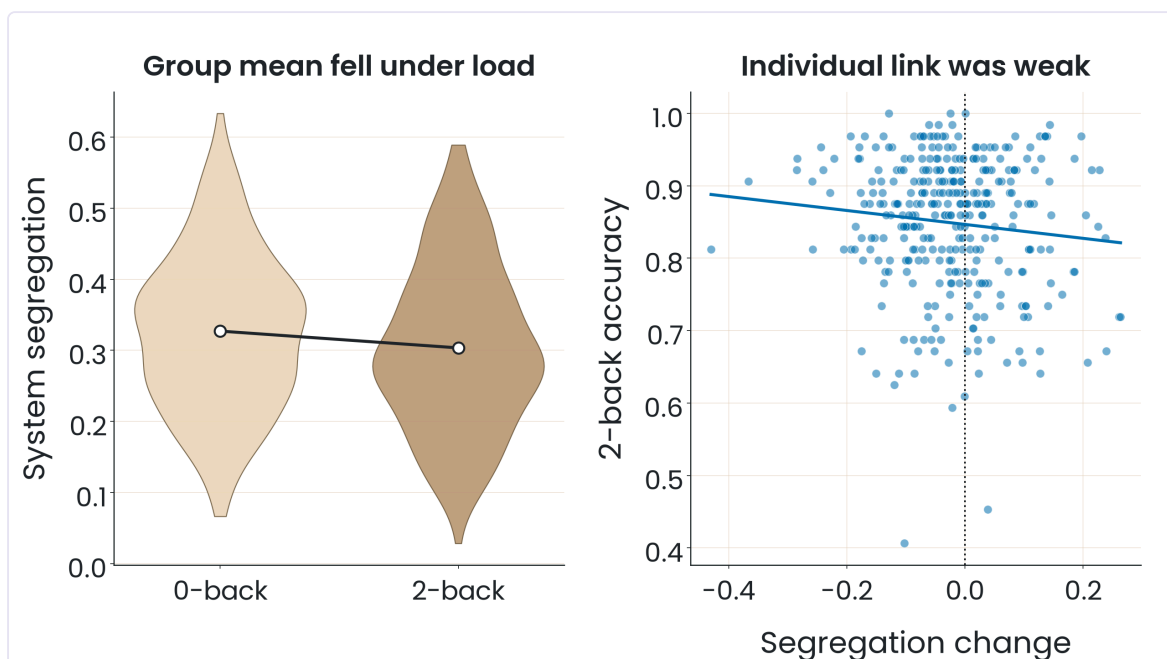
**Likely questions:** this slide is your Q&A map — keep it projected. Each column points to a backup: Refined → slides 8 and 9, Unresolved → slide 10 (future work).

**Avoid:** saying the conclusion "changed"; that a connectivity mechanism was demonstrated.

## Backups (open only if asked)

### Slide 8 — Backup: the directional result in full

**On screen:** Title "Group direction was real; the individual link was weak" · "Group mean: 0-back 0.3271 → 2-back 0.3035" · "Paired change:  $\Delta = -0.0236$ ;  $p = 3.45 \times 10^{-5}$ " · "Across participants:  $r = -0.105$ ;  $p = .054$ " · "A reliable mean shift does not establish individual predictive relevance." · Figure segregation-refinement.png .



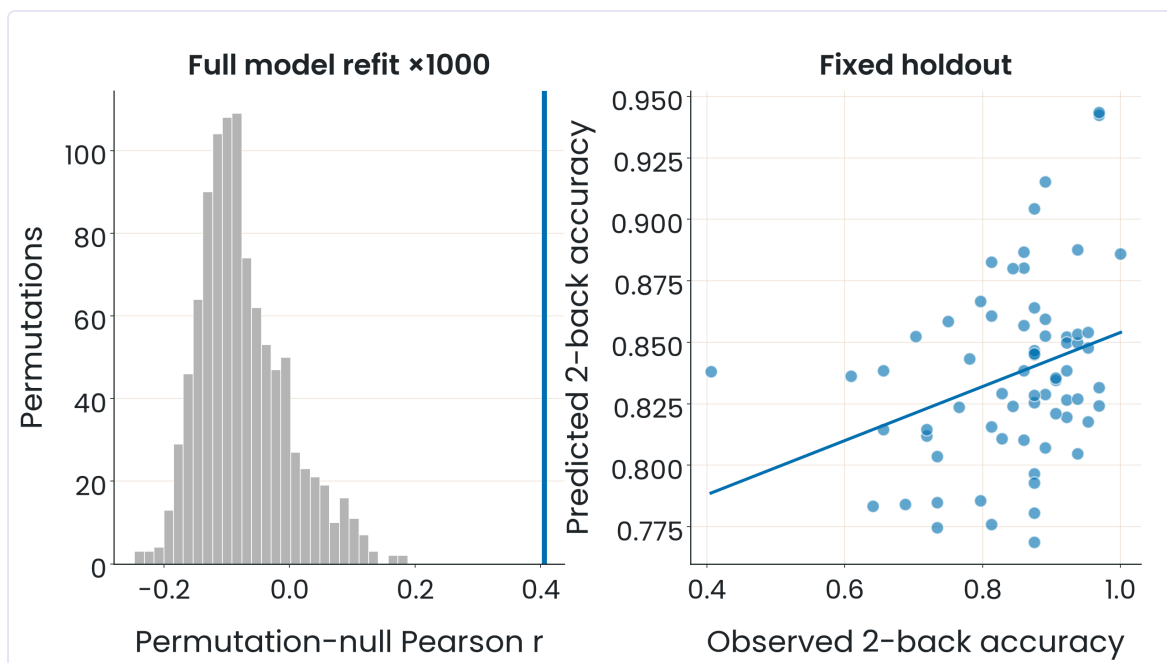
**The figure (2 panels):** left, two **violins** (0-back / 2-back) = the full segregation distribution of the 336 people, not just the mean; a line with two white dots connects the means and shows the 0.3271 → 0.3035 drop. Right, **scatter**: X = each person's segregation change, Y = their 2-back accuracy; a dotted vertical line at 0 (left = integrated more). The trend is almost flat. **Real group effect, weak individual link** — slide 6's lesson with the full picture.

**When to open:** "what does 'the directional hypothesis was refined' actually mean?".

**Note:** don't reuse the **-0.048** magnitude from the submitted abstract — that used a different atlas convention; the canonical one is -0.0236 (same sign, half the magnitude).

## Slide 9 — Backup: the checks behind the primary result

**On screen:** Title "The checks behind the primary result" · "Fixed holdout:  $r = 0.312$  in 67 unseen participants" · "Full-refit null (seed 42):  $r = 0.405$ ;  $p = 1/1001 \approx .001$ " · "B → A A-label permutation, fixed B predictions:  $p = 1/1001 \approx .001$ " · "Reconfiguration over 0-back FC:  $\Delta R^2 = +0.0344 \pm 0.0225 \rightarrow$  no clear gain" · "FC over activation:  $\Delta R^2 = -0.0030 \pm 0.0065 \rightarrow$  no clear gain" · "The permutation  $p$  belongs only to seed-42  $r = 0.405$ ; the holdout is a separate split." · Figure `null-and-holdout.png` .



**The figure (2 panels):** left, **histogram** = shuffle the performance labels 1000 times (refitting the model, fixed partition) → the distribution of "what chance looks like with this pipeline"; a vertical line marks the real  $r = 0.405$ , to the right of almost the whole cloud → only 1 of 1001 was that extreme →  $p \approx .001$ . Right, **fixed holdout**: 67 never-touched people,  $r = 0.312$ .

⚠ The one-in-a-thousand  $p$  belongs **only** to the seed-42  $r = 0.405$  — not to the 0.312 holdout nor the 0.366 repeated-CV. Three different procedures, same story, not interchangeable.

**The two  $\Delta R^2$  ("no clear gain"):** a "2-SD" heuristic (the increment is smaller than twice its own variability).

**Not a formal** superiority/equivalence test — say so if asked.

**When to open:** "how do you know it's not chance / and the holdout?".

*Extra ammunition (in `pipeline/02`, not on a slide):* we also partial out 0-back accuracy (to rule out "general ability") and repeat with **d'** (signal-detection sensitivity) in cohort B. If asked "did you control for general ability?" → yes.



## Slide 10 — Backup: future work

**On screen:** Title *"Five tests could resolve the remaining question"* · 5 items + decision criterion ("FC must add reliable held-out value beyond activation and single-condition FC.").

**The 5, what each attacks:** (1) match activation/FC dimensionality in nested CV; (2) separate coactivation from coupling with a prespecified estimator; (3) independent-site / repeat-session generalization with kinship modelled; (4) run the activation model on **resting-state** — if rest predicts as well as task, the signal is **task-independent** (runnable now: cohort B ships 4 rest runs); (5) **ST-GNN** (Goutham's proposal, already building it) — model the transition as a dynamic graph and let the model **learn** the reorganization rather than summarize it in one difference ("watch the film, not two snapshots").

**When to open:** "what next / how would you resolve the activation-vs-FC question?".

## Slide 11 — Backup: references and statistical guardrails

**On screen:** Title *"References and statistical guardrails"* · bibliography (Avery 2020, Chan 2014, Murphy 2020, Masharipov 2024, Hedge 2018, Logothetis 2008) + guardrail definitions ( $r = 0.366 \pm 0.024$  = mean  $\pm$  split SD over 20 partitions; seed-42  $r = 0.405$ ,  $p \approx .001$  = full-refit null;  $B \rightarrow A$   $r = 0.398$ , CI [0.25, 0.53] = identity-disjoint transfer).

**How to present it (Andrea):** the last presenter shows it for ~5 seconds, says "these are the references we used", and moves to questions. **Don't read it aloud.** Chan backs the segregation formula; Hedge backs Goutham's argument (slide 6); Logothetis frames the CVR limit as a **known** field limitation, not a finding of ours.

**Three numbers to memorize (they carry the whole talk):**  $0.366 \pm 0.024$  · seed-42 0.405,  $p \approx .001$  ·  $B \rightarrow A$  0.398, CI [0.25, 0.53].